

MOHAMED AGHNAOU



Software Engineer seeking a full-time position.

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Software Development Engineer, with a strong background in designing and building end-to-end software solutions. Skilled in backend and frontend development, database design, and application integration. Experienced in developing scalable, maintainable applications and working across the full software lifecycle.

Education

- 📅 2022–2025 Software Development Engineering
National Institute of Posts and Telecommunications (INPT)
- 📅 2020–2022 Preparatory Classes for Engineering Schools (MP)
Lycée Réda Slaoui - Agadir

Professional Experience

ORACLE 📅 01/2025–09/2025 📍 Casablanca, Morocco
Software Engineer – Research Assistant

- Designed and implemented a backend feature enabling users to unlink Intercompany Cross Charge Journals (ICCs) directly from the NetSuite interface.
- Ensured financial data integrity by enforcing accounting period constraints, permission rules, and guaranteeing atomicity of database operations.
- Established a comprehensive testing strategy including unit, component, and E2E tests, ensuring reliability of the feature in a production environment.

Technologies : Java, NetSuite ERP, Git, GitLab, Jira, TeamCity, Oracle Database, SuiteQL, JUnit, Mockito

Royal Air Maroc 📅 06/2024–08/2024 📍 Casablanca, Morocco
Full-Stack Engineer

- Developed a web application promoting sustainable tourism and natural travel destinations.
- Implemented interactive eco-travel exploration and booking features based on dynamic itineraries.
- Designed secure user authentication with real-time reviews.
- Integrated secure online payment processing for reservations.

Technologies : NodeJs, Express, MongoDB, Mongoose, Pug, Git, Github

Projects

Enterprise Knowledge Management AI Assistant 01/2026–3/2026

Design and development of a full-stack AI assistant enabling natural language querying of 500+ internal documents:

- Designed and implemented a multi-tenant RAG pipeline using Spring AI, OpenAI embeddings, and PGVector, covering document ingestion with Apache Tika, vectorization, and semantic search with tenant isolation.
- Developed an AI Agent dynamically orchestrating multiple tools via MCP (filesystem, external APIs, database) enabling autonomous task execution beyond document Q&A.
- Implemented real-time AI response streaming via SSE, JWT-secured REST API, and role-based access control with four distinct user roles.

Technologies: Java, Spring Boot, Spring AI, OpenAI API, PGVector, PostgreSQL, MinIO, Angular, Docker, MCP

BiblioManager — AI-Powered Library Management System 08/2025–12/2025

Developed a full-stack library management system enhanced with a Generative AI chatbot using RAG:

- Implemented an intuitive system enabling users to add, search, and track available books.
- Developed reservation management and user review features.
- Integrated a Generative AI chatbot using RAG, LangChain4j and Google Gemini API enabling natural language book search with context-aware chat memory and semantic retrieval from a live MySQL database.

Technologies : ReactJs, Bootstrap, Spring Boot, Spring Security, Spring Data JPA, LangChain4j, Google Gemini API, RAG, Vector Embeddings, MySQL, REST APIs

ReserveINPT 06/2023–08/2023

Developed a mobile reservation application for managing institutional facilities:

- Enabled users to create, modify, and cancel reservations.
- Provided administrative tools for managing student accounts and facilities.

Technologies : Java, AndroidStudio

Skills

Programming : Java, JavaScript, TypeScript, Python, Kotlin

Backend Development : JEE, Spring, Spring Boot, Spring Data, Spring Cloud, Spring Security, Keycloak, NodeJs, Express

Frontend Development : HTML, Bootstrap, Tailwind CSS, ReactJs, Angular, Next.js

Generative AI & LLMs : LLMs, Prompt Engineering, Retrieval-Augmented Generation (RAG), Multimodal AI, OpenAI API, LLaMA

AI Agents & Automation : Agentic AI, Model Context Protocol (MCP), Spring AI, Workflow Orchestration, Tool Calling, LangChain4j

DevOps : Docker, Kubernetes, Kafka, GitLab CI/CD, Jenkins

Databases : MySQL, SQL Server, MongoDB, Oracle Database, PostgreSQL

Software Architectures : Microservices, Event Driven Architecture, Event Sourcing, CQRS, Saga Pattern OAuth2.0, REST APIs, GraphQL,DDD

Languages

Arabic : Native

English : Professional

French : Fluent (DELF B2)

Soft Skills

Communication - Teamwork - Problem Solving - Adaptability - Self-learning